

VitalStep: Revitalizing Elderly Health and Connectivity with E-Square-Dance

Jun Huang*

12111119@mail.sustech.edu.cn
School of System Design and
Intelligent Manufacturing
Southern University of Science and
Technology
Shenzhen, Guangdong, China

Qianwen Tan*

12332673@mail.sustech.edu.cn
School of System Design and
Intelligent Manufacturing
Southern University of Science and
Technology
Shenzhen, Guangdong, China

Qi Fang*

12332652@mail.sustech.edu.cn
School of System Design and
Intelligent Manufacturing
Southern University of Science and
Technology
Shenzhen, Guangdong, China

ABSTRACT

In response to the challenges faced by China's aging population, including mobility restrictions, health concerns, and social isolation, calls for innovative approaches to enhance their physical and mental well-being. Square dancing, or 'Guangchang Wu', is a communal activity where Chinese elderly gather in public spaces to dance together, combining exercise with social engagement. We developed VitalStep, an online co-dancing app tailored for people aged 45-70 to participate in square dancing virtually. Through collaboration with square dance communities, we ensured our app met the needs and preferences of its target users. Early findings suggest that the app successfully motivates seniors to be more physically active and socially engaged. Our project aligns with the United Nations Sustainable Development Goals 3 (Good Health and Well-Being) and 11 (Sustainable Cities and Communities), demonstrating the potential of technology in fostering healthier, more inclusive urban environments for the middle-aged and elderly.

CCS CONCEPTS

• **Human-centered computing** → **User centered design**; *Interface design prototyping*.

KEYWORDS

aging population, square dance, digital health, social engagement, user-centered design

ACM Reference Format:

Jun Huang, Qianwen Tan, and Qi Fang. 2024. VitalStep: Revitalizing Elderly Health and Connectivity with E-Square-Dance. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3613905.3647978>

1 INTRODUCTION

The trend of population aging has become one of the most significant demographic characteristics of the first half of the 21st century.

*All authors contributed equally to this research.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CHI EA '24, May 11–16, 2024, Honolulu, HI, USA
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-0331-7/24/05
<https://doi.org/10.1145/3613905.3647978>

By 2023, the population aged over 60 is projected to increase by more than 40.8% in the next two decades[18]. China, in particular, is poised to witness a substantial demographic shift, with the elderly population expected to exceed 365 million by 2030[18]. Addressing the needs of this vast elderly population and ensuring they live their later years with dignity has become an urgent societal issue in China[15].

Regarding physical health, about a quarter of Chinese people report at least one Activity of Daily Living (ADL) difficulty by the age of 45[14], and over 40% suffer from hypertension[7]. Lack of exercise has been identified as a crucial factor[7]. In contrast, approximately two-thirds of the American population over 50 engage in moderate or higher levels of physical activity[17], while in China, about 49.4% of the elderly never exercise[3].

Notably, several studies have also linked the lack of physical activity to the prevalence of mental health issues in the middle-aged and elderly[1, 13]. While the social factors in China further complicate the psychological issues of the elderly. The migration of younger generations to urban centers and the changing dynamics of family structures have left many elderly individuals either leaving their familiar social circles or lacking companionship from their children[9], leading to increased feelings of loneliness and social isolation[14].

In this context, Square dancing ('Guangchang Wu'), a special dancing culture in China, particularly among the middle-aged and elderly, offers an innovative pathway to tackle these challenges. Square dance is a communal dance, often performed in public spaces like parks and squares, typically involves groups ranging from 10 to 100 people, providing not only physical benefits but also a means of fostering community interaction. Studies show that activities combining physical and social elements can significantly improve both physical and mental health among the elderly[6, 13].

Recognizing the intersection of these issues with the United Nations' Sustainable Development Goals, especially **SDG3: Good Health and Well-being** and **SDG11: Sustainable Cities and Communities**, our project seeks to innovate in this space. We observed local square dancing and interviewed a range of individuals, from organizers, regular participants to those hesitant to join, targeting on those aged between 45 and 70. We found that making dance learning easier and reducing social barriers are the aspects most valued by our users. Additionally, physical limitations like weather and distance were identified as key challenges in participation.

user research phase employed qualitative interviews and behavioral observations. We conducted field research at a local square in Shenzhen, known for its vibrant square dancing atmosphere. Eight middle-aged and older individuals were interviewed, including three males and five females, aged between 40 and 65. Additionally, we observed and recorded the behavior and activity of local residents participating in square dancing for eight hours.

2 RELATED WORK

While in the realm of near-elderly entertainment and square dancing, platforms like Douyin¹ (China's domestic version of TikTok) and Kuaishou² have made significant strides. While excelling in content delivery and popularizing square dance content among the them, they primarily function as platforms for passive learning. Features like live streaming offer real-time viewing experiences but lack the interactivity essential for active participation and learning. Furthermore, these platforms, despite their success in content reach, do not provide functionalities like step-by-step dance breakdowns or personalized feedback, which are vital for the users above 45 to learn new dances effectively.

[illegible]

Figure 1: The Affinity Diagram is derived from interview.

Scenario		LI 60-65 years old She needs to use the mobile app to find the suitable dance to learn and share it with her dance group friends.				
Phase		Search	Preview and Mark	learn the video	Video Record	Share
Actions	Search for dance that are appropriate difficult and enjoyable to watch. Research about: - dance video publishers - different types of dances - song linkages	Preview the video: - drag the progress bar - adjust playback speed	Choose a video and follow along with a video to learn or exercise - click to try - click to like - enter video playback speed - make playlist - click video to loop - learn to upload the video explanation	Take videos of dancing then post - Find the suitable angle to post - Add timeline tagging with shot	- find the share button and share to WhatsApp moment or WhatsApp group and friends	
	Do research through: - app/YouTube/TikTok/Instagram - home page recommendations - Keyword/Hashtag Search - Category Search					
Pain Points	inability to use text input properly The interface is complicated.	- want to understand what the icons meaning - Don't know how to get to the homepage page	- Always have to drag the progress bar over and over again - Some movements are difficult to understand and memorize - Want to know if my movements are standardized	- Can't find the right camera - The video that is not aesthetically pleasing enough	- Sharing function isn't - Don't know how to share to another software	
Emotions						
Insights/Opportunities	- Support voice search - Provide advice on using category search - Labeling the difficulty of dance	- Simple and obvious favorite buttons and pages	- Provide actual descriptions and audio-explanation of dance - Simple 1/2 screen costing function - Real-time display of individual dance moves	- Recording while learning to dance - Record videos showing actual backgrounds or outdoors - Automatic videos editing and beautification	- Incorporate the Dance Club/Cost feature into the dance app - Clear and simple sharing process	

3 USER RESEARCH

In analyzing our interview data, we transcribed the recordings into text and anonymized all details to ensure privacy. We then selected 47 key statements to create our affinity diagram. To enhance the diagram's effectiveness in defining design implications,

¹<https://www.douyin.com>
²<https://www.kuaishou.com>
³<https://www.tangdou.com>
⁴<https://justdancenow.com>
⁵<https://www.nintendo.co.uk/Wii>

we employed Cohen's Kappa for a quantitative assessment of our categorization process[8]. We involved two external individuals with an understanding of square dancing to serve as independent "coders".

After two rounds of evaluations and refinements, we reached an average Cohen's Kappa score of 0.915, with an agreement rate of 97.89%. This high concordance rate shaped the construction of the affinity diagram shown in Figure 1.

From the ideas we summarized, we finally identified several key user insights:

- (1) Square dancing fosters positive social interaction and self-expression among middle-aged and older adults, enhancing their social network and well-being. A 55-year-old female interviewee mentioned: "I came here from hometown to help my daughter take care of kids, and when I first arrived in Shenzhen, I felt very unfamiliar and lonely. Later, I joined the dance group in the community, where I met many new friends. This healed me."
- (2) Some users desire social interaction but lack confidence. Joining large offline square dance groups is a bit of a threshold for older people with no dance experience. One respondent said: "I usually just watch from the sidelines, I can't dance well, I feel out of place and I'm afraid of being laughed at by others."
- (3) The growing popularity of square dancing across age brackets creates common ground for everyone, and often can strengthen family bonds. In our observations, some larger dance groups had nearly equal proportions of young and older participants. However, one young dancer we interviewed expressed a desire to join the square dancing group but was apprehensive about potential rejection and judgment from others.
- (4) Many interviewees find it challenging to proficiently use software, resulting in that they can only learn square dancing offline. Also, users have to switch between apps when learning and sharing square dancing videos now. The lack of uniformity of dance contents increases the learning costs for the elderly users. A dance group leader mentioned that they typically find suitable dances on Douyin (TikTok), then learn the steps on Tangdou Square Dance App, and share them on WeChat for members who are not familiar with using software.

4 DESIGN IMPLICATIONS

Based on the preliminary research results, we created user personas, categorizing users into two groups: Home Fitness Enthusiasts and Square Dance Socializers. The first group of users shows a willingness to learn square dancing or habitually engage in home fitness activities. For them, square dancing is a way to improve health and relaxation, seeking convenience and enjoyment in home workouts. On the other hand, The second group enjoys participating in square dance groups to combine physical fitness with socializing, making friends and maintaining positive social relationships.

Our design goals are as follows:

- Design a simple, easy-to-use interface for the square dance app, reducing the cognitive load for middle-aged and older adults when using software.
- Integrate features that commonly used in dance software, providing more comprehensive and intuitive dance guidance. Reduce operational costs associated with switching between different software platforms.
- Combine online learning with offline dancing, promoting online dance socialization among user groups of different age ranges through various social interaction modes.

5 DESIGN PROCESS

5.1 Proposed Concept

Our team initiated a brainstorming to explore various digital solutions. We listed and evaluated a range of potential functional requirements and conducted a series of trade-offs. Through the analysis, we excluded complex gamification elements, such as live competitions and leveling systems. Instead, we emphasized a user-friendly interface, clear instructional content, and community features that encourage interaction and mutual support among users. Additionally, it should be responsible for transforming the time and space limitations of offline square dancing into the collective and distributed characteristics of online dance, so as to reduce the participation threshold of middle-aged and elderly users.

We proposed VitalStep, an application designed to facilitate online square dancing. The conceptual framework of VitalStep was three main functionalities: individual dance learning, online square dancing, and square dance community.

First, VitalStep provides low cognitive load features such as real-time feedback and steps breakdowns to assist in the learning process. Second, the online square dancing function aims to alleviate the psychological and physical barriers that the aged above 45 may face in socializing, such as the stress of outdoor socialization or weather constraints. Finally, the square dance community facilitates the formation of a sustainable interest community among dancers, range from local dance event postings to experiences exchange from different cities.

5.2 Preliminary Prototype

Based on the conceptual outcomes, we drafted a preliminary prototype, shown in Figure 3. These preliminary prototype outlined the selection process of dances learning, interfaces for online square dance, and community interfaces for social sharing and interaction. Then, we sought feedback from professionals with these preliminary prototype. The feedback indicated that we had to decide whether real-time feedback for dancers should be represented through live camera images or virtual avatars. Also, they suggested that the feedback rhythm had to align with the cognitive abilities of our users.

Addressing these concerns, we designed an experiment. For the purpose of quick testing, in this stage, we decided to make re-recorded videos for non-real-time motion capture, and used Blender 3.6 for skeletal motion rendering of the avatar. We then asked a 52-year-old dancer following a tutorial from Tangdou Square Dance and recorded it. Finally, two sets of videos were produced (see Figure 4) contrasting "live human versus virtual avatars" and "fast-paced



Figure 3: Preliminary prototype of the overall framework of our potential APP.

versus slow-paced feedback", along with a control group, resulting in five videos in total.

Focus on these two questions, we conducted Likert scale assessments on 32 users aged 40 and above. The results showed average scores of 3.50, leaning towards a preference for virtual avatars; and 3.57, favoring more intensive feedback. Furthermore, Wilcoxon signed-rank tests yielded p-values of 0.034 and 0.014, respectively, both below 0.05, indicating statistical significance. Based on these, we opted for incorporating virtual avatars and intensive feedback.



Figure 4: Four screenshots from different assessment videos.

5.3 Wireframe Prototype

We developed our wireframe using the insights generated from our preliminary prototyping phase, shown in the Figure 5. These wireframe were then subjected to two rounds of design critiques. The first critique session involved three potential users from local square dancing communities, and the second one was conducted within our design team.

The critiques highlighted an urgent need for improvement in the scalability of fonts and icons to ensure that older adults can easily read every piece of information on their mobile screens. Additionally, two elderly participants specifically mentioned their preference for simplicity on each page, with spacious layouts; one of them commented, "We often start frowning when faced with overly congested interfaces."

While there was a favorable opinion towards the use of virtual avatars, they did not see themselves enjoying the process of extensive customization. Instead, they indicated that a selection of preset avatars would be more user-friendly.

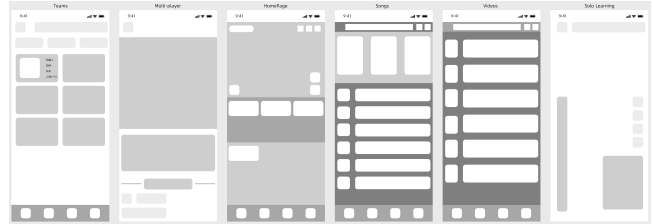


Figure 5: Wireframe of six main interfaces of VitalStep.

5.4 High-fidelity Prototype

Based on the insights from the design critiques, we completed the creation of our high-fidelity prototype (see in Figure 6). We enlarged the fonts universally from 12pt to 20pt and simplified information on each page.

We also observed that middle-aged and older adults tend to choose whether to participate or rest on the sidelines during square dancing, depending on the speed and type of the songs. This observation inspired us to show the current song in the VitalStep room card and add indications related to the rhythm speed in the solo learning module.

We distributed our high-fidelity prototype as images in the square for a broad and random collection of evaluations from various square dance participants. The comprehensive results showed widespread praise for the functionality of VitalStep. Many users drew direct comparisons to the Tangdou Square Dance app, which they commonly use, noting that online square dance would provide them with the opportunity to dance anytime and anywhere.

Nevertheless, we also received valuable insights for revision. A leader and organizer of a dance team consisting of more than 40 members mentioned that, when choosing dances, the most critical aspect they consider is the instructor in the video. Often, the dances learned by a team are selected from a few internet-famous instructors they trust. Also, she felt that the displayed space on mobile phones was too small, thus TV casting was one of the most desired features to be included.

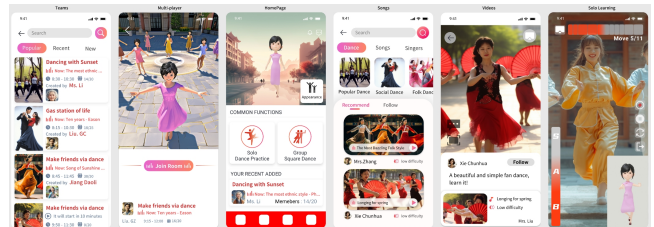


Figure 6: Some examples of high-fidelity prototype corresponding to the wireframe.

6 THE LATEST DESIGN

Including all the feedback throughout our design process, we crafted the latest iteration of VitalStep.

6.1 Solo Dance Practice Room

Users can directly access the Solo Dance Practice Room page from the homepage. In this task flow, shown in the Figure 7. a), they can filter dance by dance/song genre or dance instructors. To minimize page transitions, corresponding dances will be displayed immediately below the categories. Each dance preview card includes the instructor, song title, and difficulty level – the three most critical factors for middle-aged and older users when deciding to dance or not.

After choosing a song, users can engage in social actions such as commenting, following, and sharing. By pressing the fullscreen button, users begin their dance practice. During practice, they will receive feedback on their move accuracy every beat, displayed as a dynamic score column on the left. Additionally, the top shows current progress and remaining time in the form of dance steps breakdown. Users can record, pause, repeat, or exit at any time in this page.

A TV casting button is specially put on the upper left corner. Once cast to the TV, as shown in Figure 7. b), the interface allows for a larger display of dancers and textual posture feedback, leveraging the larger screen space.

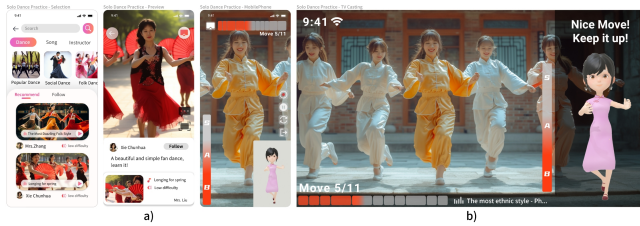


Figure 7: Solo Dance Practice: a) Dance Selection and Learning, b) TV Casting.

6.2 Online Square Dance Room

For the Online Square Dance Room, considering the characteristic social circles of square dancing, VitalStep displays on the homepage that the online square dance rooms user has recently joined. Additionally, users can navigate to square dance page from the homepage, which lists various 'live rooms'. The live room cards highlight preview images, team names, and leaders for quick peek, with details like song, duration, and participant count displayed secondarily.

Upon selecting a room, users can watch a live preview before deciding to join. Once joining, their real-time motion-captured virtual avatar integrates into the dance group. Users can switch views by tapping the window in the lower right corner, as shown in Figure 8. a). Social features such as short video sharing and recording are readily available. Also, similar to the Solo Dance Practice, we provide a TV Casting button on the top.

Moreover, every user can create their own room. By simply clicking the plus sign in the lower right corner of the room selection

page, they are directed to the room creation page, as illustrated in Figure 8. b).

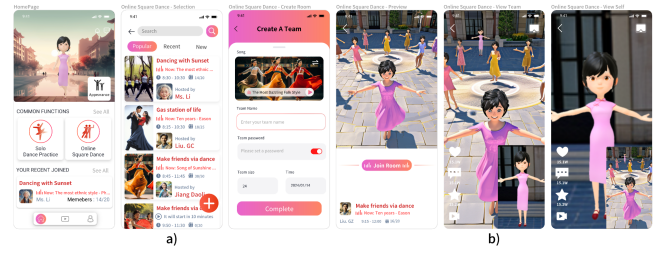


Figure 8: Online Square Dance: a) Room Selection, b) Online Square Dance Room.

6.3 Square Dance Community

In the Square Dance Community section, we added a dedicated button in the bottom bar, making it a standalone feature distinct from the previous dancing features. In both Solo Dance Practice and Online Square Dance rooms, we've introduced a recording button in the interface. Now, after each recording, VitalStep offers the option to upload the video to the community. Once uploaded, as shown in Figure 9. a), it appears in the community's waterfall feed. Users can engage in basic social interactions such as sharing and commenting here, similar to short video apps.

Specially, users can directly jump to the Solo Dance Practice to learn the steps or join a dance group in the Online Square Dance room, by clicking the song or dance team information displayed on the bottom. This feature was inspired by our research findings, where many dance team leaders mentioned their current workflow involves seeing a dance they like on Douyin (TikTok) and then looking for steps tutorials on apps like TangDou. We seamlessly integrated this process, offering a distinctive and user-centric solution.

Additionally, users can access their personal profiles through the bottom bar. The profile page includes an overview of their account, recent works, social interactions, and comments, as shown in Figure 9. b).

7 EVALUATION

We evaluated the latest VitalStep to see if it is easy for its target users to use (shown in Figure 10). The participant demographics included six females and two males, with five being new participants and three from previous research, all between the ages of 36 to 58. The detailed procedure is as followed. Initially, participants were briefed on the app's purpose and functionalities, with a particular focus on the three main modules discussed earlier. Following this, they were presented with our interface design prototype and were tasked to independently complete four specific tasks from the landing page: (A) starting a solo dance, (B) starting an online square dance, (C) recording and sharing a video to the community, and (D) watching a community video and then navigating to the related learning page. The ability to complete each task unaided was recorded. Noting that time was not the primary metric, yet any task that was not completed within 40 seconds was marked as unsuccessful.

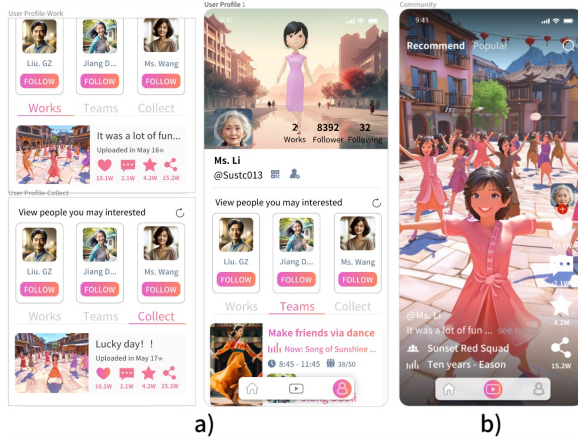


Figure 9: Square Dance Community: a) User Profile, b) Videos Feed.



Figure 10: Photo taken when we did evaluation.

The results, as shown in Table 1, demonstrate a task completion rate of 93.75%, with the majority of participants able to independently navigate the prototype and complete the tasks. In the subsequent interviews, seven participants expressed a keen desire to download the app immediately. The most praised feature was the ability to seamlessly transition from viewing a dance in the community to learning the steps or joining the online dance. One evaluator remarked, “This is an eye-catching and much-needed feature. I would download the app just for this.”

Table 1: Evaluation Task Completion by Participants. ('Y' for a successful completion and 'N' for a task not completed)

	P1	P2	P3	P4	P5	P6	P7	P8
Task A	Y	Y	Y	Y	Y	Y	Y	Y
Task B	Y	Y	Y	Y	Y	Y	Y	Y
Task C	Y	Y	Y	N	Y	Y	Y	Y
Task D	Y	Y	Y	N	Y	Y	Y	Y

8 FUTURE WORK

Our evaluation results show that VitalStep has fulfilled some latent yet urgent needs of our users. Some evaluators expressed a desire for an immediate real-time experience of learning dance with the aid

of virtual avatars. A key technological aspect, as becomes evident, is the utilization of smartphone hardware to achieve monocular camera-based motion capture, real-time rendering synchronization, and movement recognition/classification. We previously demonstrated these capabilities using pre-recorded videos and Blender 3.6 on a Windows 11 platform. However, for ease of deployment on mobile devices, we transitioned to using the Unity engine instead of Blender due to its widespread deployment capabilities on Android and iOS platforms.

Furthermore, in our quest to implement real-time motion capture, we experimented with Microsoft’s Kinect 2⁶ and Apple’s ARKit 6⁷. However, both were found to be limited to specific hardware from their respective companies. Consequently, we opted for Google’s recently open-sourced MediaPipe solutions⁸. MediaPipe not only allows for effortless deployment across Android, iOS, and Web platforms but also integrates lightweight machine learning features[11]. Its end-to-end optimization frees us from the constraints of hardware computational power. Recent studies have also demonstrated that MediaPipe can recognize a variety of yoga poses under modest computational loads[5].

Our idea has been recognized and supported with a grant of 40,000 RMB from the 2024 "Climbing Program" in Guangdong Province, China. Our extended plan also includes integrating this project with some hardware like flexible wearable functional electrical stimulation devices, initially expanding into community rehabilitation applications for stroke patients.

9 CONCLUSION

In this project, we started from the increasing physical and psychological issues within China’s aging population, uncovering square dancing as a culturally significant potential solution. Our multiple field studies revealed many unexpected insights, particularly valuable given the significant age difference between our user group and our team. Through stages of Preliminary Prototype, Wireframe, High-Fidelity Prototype, and the Latest Iteration, each followed by seeking new modifications from users, we accomplished a user-centered design. VitalStep lowers the barrier for users aged 45-70 to engage in square dancing and establishes a social community centered around it. We believe VitalStep will contribute to achieving the SDG 3 (Good Health and Well-being) and 11 (Sustainable Cities and Communities).

ACKNOWLEDGMENTS

We extend our deepest gratitude to the square dance organizations in Longhua and Longgang Districts of Shenzhen, China. Their friendly and enthusiastic cooperation were pivotal to the success of our project. Special thanks to Ms. Peng and her square dance team for providing invaluable insights into the current state of square dance in China.

We would also like to express our heartfelt appreciation to our supervisor, Professor Ziqian, Bai. Without her support and guidance, this project would not have been possible from the outset.

⁶<https://www.microsoft.com/download/details.aspx?id=44561>

⁷<https://developer.apple.com/augmented-reality/arkit/>

⁸<https://developers.google.com/mediapipe>

This work was supported by Special Funds for the Cultivation of Guangdong College Students' Scientific and Technological Innovation ("Climbing Program" Special Funds): pdjh2024c11602 and pdjh2024c11604.

REFERENCES

- [1] Marybeth Brown, David R Sinacore, Ali A Ehsani, Ellen F Binder, John O Holloszy, and Wendy M Kohrt. 2000. Low-intensity exercise as a modifier of physical frailty in older adults. *Archives of physical medicine and rehabilitation* 81, 7 (2000), 960–965. <https://doi.org/10.1053/apmr.2000.4425>
- [2] Ying-Yu Chao, Yvonne K Scherer, and Carolyn A Montgomery. 2015. Effects of using Nintendo Wii™ exergames in older adults: a review of the literature. *Journal of aging and health* 27, 3 (2015), 379–402. <https://doi.org/10.1177/0898264314551171>
- [3] J Dang. 2018. Survey Report on the Living Conditions of the Elderly in Urban and Rural Areas of China. *Social Sciences Literature Press: Beijing, China* (2018).
- [4] Susan M Ferreira, Sergio Sayago, and Josep Blat. 2017. Older people's production and appropriation of digital videos: an ethnographic study. *Behaviour & Information Technology* 36, 6 (2017), 557–574. <https://doi.org/10.1080/0144929X.2016.1265150>
- [5] Shubham Garg, Aman Saxena, and Richa Gupta. 2022. Yoga pose classification: a CNN and MediaPipe inspired deep learning approach for real-world application. *Journal of Ambient Intelligence and Humanized Computing* (2022), 1–12. <https://doi.org/10.1007/s12652-022-03910-0>
- [6] Kathrin Gerling, Mo Ray, Vero Vanden Abeele, and Adam B. Evans. 2020. Critical Reflections on Technology to Support Physical Activity among Older Adults: An Exploration of Leading HCI Venues. *ACM Trans. Access. Comput.* 13, 1, Article 1 (apr 2020), 23 pages. <https://doi.org/10.1145/3374660>
- [7] Dongfeng Gu, Kristi Reynolds, Xigui Wu, Jing Chen, Xiufang Duan, Paul Muntner, Guanyong Huang, Robert F Reynolds, Shaoyong Su, Paul K Whelton, et al. 2002. Prevalence, awareness, treatment, and control of hypertension in China. *Hypertension* 40, 6 (2002), 920–927. <https://doi.org/10.1161/01.HYP.0000040263.94619.D5>
- [8] Jonathan Lazar, Jinjuan Heidi Feng, and Harry Hochheiser. 2017. *Research methods in human-computer interaction*. Morgan Kaufmann. 317–320 pages. Section 11.4.3.2 Reliability.
- [9] Xiaoyan Lei, John Strauss, Meng Tian, and Yaohui Zhao. 2015. Living arrangements of the elderly in China: Evidence from the CHARLS national baseline. *China economic journal* 8, 3 (2015), 191–214. <https://doi.org/10.1080/17538963.2015.1102473>
- [10] Stephen Lindsay, Daniel Jackson, Guy Schofield, and Patrick Olivier. 2012. Engaging older people using participatory design. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Austin, Texas, USA) (CHI '12). Association for Computing Machinery, New York, NY, USA, 1199–1208. <https://doi.org/10.1145/2207676.2208570>
- [11] Camillo Lugesia, Jiuqiang Tang, Hadon Nash, Chris McClanahan, Esha Uboweja, Michael Hays, Fan Zhang, Chuo-Ling Chang, Ming Guang Yong, Juhyun Lee, et al. 2019. Mediapipe: A framework for building perception pipelines. *arXiv preprint arXiv:1906.08172* (2019). <https://doi.org/10.1101/1906.08172>
- [12] Donald A. Norman and Stephen W. Draper. 1986. *User Centered System Design; New Perspectives on Human-Computer Interaction*. L. Erlbaum Associates Inc., USA.
- [13] Pavitra Paul, Mihran Hakobyan, and Hannu Valtonen. 2016. The association between self-perceived health status and satisfaction with healthcare services: Evidence from Armenia. *BMC health services research* 16 (2016), 1–13. <https://doi.org/10.1186/s12913-016-1309-6>
- [14] James P Smith, John Strauss, and Yaohui Zhao. 2014. Healthy aging in China. *The journal of the economics of ageing* 4 (2014), 37–43. <https://doi.org/10.1016/j.jeoa.2014.08.006>
- [15] Tencent ISUX. 2023. Exploration of Digital Entertainment Experience Design for the Older Age Group. <https://topd.tencent.com/detail/449806>. Accessed: 2023-12-27.
- [16] Yin-Leng Theng, Puay Hoe Chua, and Tan Phat Pham. 2012. Wii as entertainment and socialisation aids for mental and social health of the elderly. In *CHI '12 Extended Abstracts on Human Factors in Computing Systems* (Austin, Texas, USA) (CHI EA '12). Association for Computing Machinery, New York, NY, USA, 691–702. <https://doi.org/10.1145/2212776.2212840>
- [17] Nick Townsend, Kremlin Wickramasinghe, Julianne Williams, Prachi Bhatnagar, and Mike Rayner. 2015. Physical activity statistics 2015. (2015).
- [18] United Nations Department of Economic and Social Affairs, Population Division. 2019. World Population Prospects 2019. <https://population.un.org/wpp/>. Accessed: 2023-12-15.
- [19] Jenny Waycott, Frank Vetere, Sonja Pedell, Lars Kulik, Elizabeth Ozanne, Alan Gruner, and John Downs. 2013. Older adults as digital content producers. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Paris, France) (CHI '13). Association for Computing Machinery, New York, NY, USA, 39–48. <https://doi.org/10.1145/2470654.2470662>
- [20] Wei Zhao, Ryan M. Kelly, Melissa J. Rogerson, and Jenny Waycott. 2023. Older Adults Using Technology for Meaningful Activities During COVID-19: An Analysis Through the Lens of Self-Determination Theory. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI '23). Association for Computing Machinery, New York, NY, USA, Article 332, 17 pages. <https://doi.org/10.1145/3544548.3580839>